

Predictive Analytics and Forecasting for Construction Projects

Abstract:

Overview: The **Predictive Analytics and Forecasting for Construction Projects** web application is developed to streamline construction project management, enabling effective task tracking, resource allocation, and data-driven decision-making. Using Flask, SQLAlchemy, and machine learning models, this platform supports managing users, projects, tasks, progress updates, and forecasting key project metrics like completion time, additional workers, and materials.

Key Features:

1. **User Authentication & Management:** Users can securely register and log in, with password hashing for authentication. Admins can manage user roles, including assigning admin privileges, blocking/unblocking accounts, and deleting users.
2. **Project & Task Management:** Admin users can create and manage construction projects, defining project names, start and end dates, and statuses. Tasks are linked to projects, with progress tracking for completion, delays, and notes.
3. **Progress Tracking:** Task progress is updated by users, including completion percentages, delay days, and notes. The overall project status is dynamically updated based on task progress.
4. **Admin Panel:** Admin users can manage other users, monitor project statuses, and perform actions like making users admins, blocking/unblocking, and deleting accounts.
5. **File Import for Project Data:** Admins can import project data from Excel or CSV files, which are processed to automatically create new projects and tasks.
6. **Machine Learning Integration:** The application incorporates machine learning models to predict key project metrics, such as the estimated completion time, required number of workers, and additional materials needed. These forecasts help project managers plan more effectively and allocate resources efficiently.
7. **Data Visualizations:** The application generates comprehensive reports and visualizations for project progress, helping project managers make informed decisions and forecast potential challenges.

Technologies Used:

- **Backend Framework:** Flask
- **Database:** SQLAlchemy (SQLite)
- **Machine Learning:** Joblib (for pre-trained models)
- **Frontend:** HTML, CSS, Jinja2
- **Security:** Werkzeug (password hashing), Flask-Login (user session management)

Conclusion: The **Predictive Analytics and Forecasting for Construction Projects** web application provides a comprehensive tool for managing construction projects. With features like user

management, project and task tracking, machine learning-powered forecasts, and data visualizations, it is an invaluable resource for project managers aiming to optimize workflows, meet deadlines, and manage resources effectively.